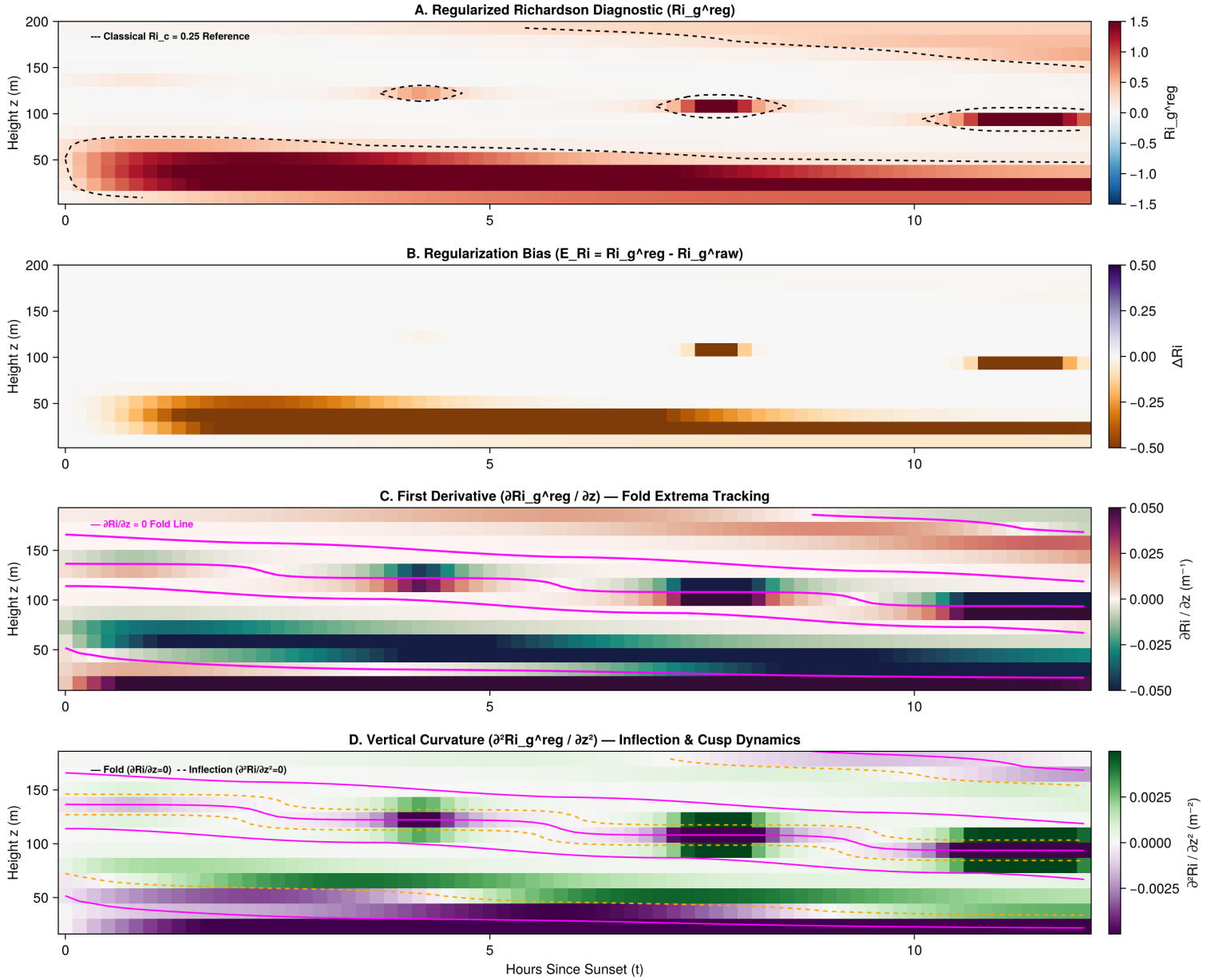


Nocturnal SBL: Gradient Richardson Number Topology, Fold Lines & Curvature



Dataset: 15-level idealized sparse SBL grid | Cusp Catastrophes occur at intersections of Fold (Magenta) and Inflection (Orange) lines.

Diagnostic Overview: Idealized Nocturnal Stable Boundary Layer (SBL)

This four-panel diagnostic suite illustrates the spatial-temporal evolution of the gradient Richardson number (Ri_g), its visualization regularization distortion, and its higher-order vertical topology across a synthetic 12-hour nocturnal cycle ($0 \leq t \leq 12$ h) sampled on a 15-level idealized tower grid ($2 \leq z \leq 200$ m, $\Delta z \approx 14.1$ m).

The underlying flow is driven by two coupled physical mechanisms: progressive surface radiative cooling producing a surface-based potential temperature inversion ($h_\theta = 45$ m), and a descending Low-Level Jet (LLJ) in the u -wind component (core descending from 140 m to 90 m with speeds peaking at 12 m s^{-1}) combined with Ekman turning in v .

Panel-by-Panel Scientific Breakdown

- **Panel A: Regularized Richardson Field (Ri_g^{reg})**

Displays the bounded diagnostic state variable mapped smoothly into $[-1.5, 1.5]$. Strong sub-jet shear production maintains a low Richardson number ($Ri < 0.25$) near the surface and directly beneath the descending jet core. Above the jet core, buoyancy suppression dominates, producing a supercritical laminar layer ($Ri > 0.25$). The dashed black contour indicates the classical empirical transition boundary ($Ri_c = 0.25$).

- **Panel B: Regularization Bias ($E_{Ri} = Ri_g^{\text{reg}} - Ri_g^{\text{raw}}$)**

Quantifies the explicit mathematical distortion introduced by the visualization transform $R_b \tanh(Ri^{\text{raw}}/R_b)$ and shear floor S_{min}^2 . Bias is zero across active mixing and weakly stable zones ($Ri \approx 0$), but becomes large negative ($E_{Ri} \ll 0$) in weak-shear, hyper-stable aloft layers where $Ri_g^{\text{raw}} \gg R_b$, explicitly isolating regularizer artifacts from physical dynamics.

- **Panel C: First Spatial Derivative ($\partial Ri_g^{\text{reg}}/\partial z$) – Fold Locus**

Maps the rate of vertical change in stability. Superimposed solid magenta contours locate the fold locus ($\partial Ri_g/\partial z = 0$), tracking the exact spatial-temporal trajectory of Richardson number local minima (shear maxima under the LLJ) and local maxima (buoyancy-dominated regions above the jet core).

- **Panel D: Vertical Curvature ($\partial^2 Ri_g^{\text{reg}}/\partial z^2$) – Inflection & Cusp Dynamics**

Details profile concavity and boundary layer layering. Dashed orange contours highlight profile inflection points ($\partial^2 Ri_g/\partial z^2 = 0$). Intersections between the solid magenta fold lines ($\partial Ri/\partial z = 0$) and dashed orange inflection lines ($\partial^2 Ri/\partial z^2 = 0$) mark geometric cusp catastrophe points where local stratification folds nucleate or collapse over the nocturnal cycle.